

# Admissible Sets for Opportunity Exposure in Privacy-Coarsened Ride-Pooling Records

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## ABSTRACT

Ride-pooling algorithms can increase feasible matches while changing which neighborhoods enter the resulting sharing opportunities. Public Chicago records retain trip-level sharing authorization, match outcomes, and pooled chain size but suppress the group identifiers needed to measure that composition. Our target is neighborhood-level *opportunity exposure*, not verified co-presence, individual socioeconomic status, or an echo chamber. We introduce BOUNDPOOL, which screens physically plausible candidate edges, learns compatibility scores from node-level match indicators through a multiple-instance noisy-OR model, and optimizes mixing statistics over every exact-cover pairing consistent with the released records. The output is an admissible interval rather than an imputed co-rider graph. In a locked Chicago-coordinate synthetic benchmark, the primary geography-equality-free score narrows the untrimmed interval by 58.6% at  $\rho = 0.95$  while retaining the hidden pairing; a geography-equality diagnostic looks sharper but excludes truth at the same score floor, revealing proxy-induced false precision. Method validation and the pre-policy feasibility audit are complete. Complete-day Chicago structural and causal estimates for the January 2026 policy change are reserved for the production data run. AI is therefore a model-indexed tightening device; the score-free physical range remains the coverage reference.

## CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Information systems** → *Geographic information systems*; • **Applied computing** → *Transportation*.

## KEYWORDS

ride pooling, weak supervision, partial identification, exact cover, urban mobility, socioeconomic exposure, privacy-coarsened data

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## 1 INTRODUCTION

Urban pooling platforms use prediction and combinatorial optimization to decide which requests can share a vehicle. A thicker pool may create more cross-neighborhood opportunities, or it may add matches mainly within already similar parts of a city. Thickness need not even raise aggregate matching when search frictions grow with market size [10], and ride-sharing density economies can distribute service access unevenly across space [9]. Our outcome is different from both match rate and access. The scientific

question is whether additional feasible matches broaden or narrow the socioeconomic composition of opportunity exposure.

The public observation operator makes that question nonstandard. Chicago’s Transportation Network Provider records report sharing authorization, realized match status, and trips\_pooled for each trip, but suppress the service-chain identifier, vehicle identifier, provider, and stable rider identity. Declaring spatiotemporally nearby rows to be co-riders would turn privacy coarsening into false certainty. Dropping the relational object would leave a node-level match-rate regression that cannot measure exposure. We take a third route: characterize the set of groupings consistent with the released records and propagate the remaining ambiguity into a service-chain opportunity estimand, not a claim of verified co-presence.

BOUNDPOOL performs three operations. It first screens candidate edges using declared temporal, origin–destination, and directional constraints. It then learns compatibility scores from node-level match labels with a noisy-OR likelihood; no pair label is created, and an edge score is not a posterior probability that two riders shared a vehicle. Finally, it minimizes and maximizes the mixing statistic over globally feasible exact-cover pairings. The untrimmed physical set is the coverage reference. A score-retention floor is a model-indexed sensitivity restriction, not a confidence interval.

Completed evidence establishes the method’s boundary, not a Chicago policy sign. On the locked known-truth generator, the candidate graph covers all hidden pairs, the primary score tightens the interval while retaining truth, and a diagnostic score with geography-equality features creates false precision by excluding truth at a stringent retention level. The production design is locked to complete Chicago market days and the January 6, 2026 policy change; the prefix extract supplies only a schema and feasibility audit. Complete-day latent-group estimation, exposure intervals, and policy signs are reserved for the production extract and are not claimed in this draft.

The paper makes four contributions. First, it formulates suppressed pooling groups as structured missing relations rather than a clustering target. Second, it maps node-level match outcomes into edge scores without manufacturing pair labels. Third, it reports optimization-based admissible intervals for opportunity exposure and audits the coverage cost of model-based tightening. Fourth, it provides a reproducible, human-experiment-free policy design whose data, estimand, diagnostics, and claim gates are fixed in advance; this is a design contribution, not a policy conclusion.

## 2 RELATED WORK AND POSITIONING

*Forward allocation is not our estimand.* Shareability and dynamic-pooling models begin with observed requests and ask how a platform should allocate them. Santi et al. connect pairwise feasibility

to maximum matching [17]; Alonso-Mora et al. enumerate feasible groups for online trip-vehicle assignment [3]; and Ozkan and Ward study forward-looking matching with heterogeneous arrivals and patience [14]. Related operations work shows that market thickness can reduce matching under search frictions [10] and that ride-sharing density economies can produce spatially unequal access [9]. We do not propose another allocation policy or assume that thickness improves matching. We solve the inverse problem created when request outcomes remain public but the realized grouping is suppressed.

*Node prediction is not our contribution.* Fair ride-pooling objectives redistribute service or driver income in an observed matching problem [15]. Chicago studies establish that pooling demand and match outcomes vary systematically across trips and neighborhoods. Soria and Stathopoulos relate disadvantage to pooled and solo demand in 127 million trips [19]; Taiebat, Amini, and Xu predict sharing requests and successful matching [20]; Mucci and Erhardt separate the sharing request from the subsequent match outcome [12]; and Sebt and Chen combine spatiotemporal clustering with match classifiers [18]. These papers establish that node behavior is predictable. A node classifier by itself is not a contribution here. Our contribution begins when those outcomes must be reconciled with a globally feasible but unobserved pairing and their uncertainty carried into exposure.

*Policy volume effects are not exposure effects.* Chicago's differentiated transportation-network-company tax already has a causal literature. Abkarian, Hegde, and Mahmassani report a shift toward shared trips after the January 2020 differential against a declining baseline [2]; Zheng et al. find lower overall ridership, growth in shared trips, decline in solo trips, and spatial heterogeneity [23]. Those results concern counts and shares. Our planned intention-to-treat analysis asks whether a policy-induced change in the authorized pool changes the admissible composition of cross-neighborhood opportunities. Because the groups are suppressed, that outcome must be optimized over feasible pairings rather than computed from a preferred graph.

*Opportunity exposure is not social contact.* Carpooling has been related to cross-racial relational costs [4], while mobility studies distinguish residential location from the composition of visited places [11, 22]. Later work constructs person-to-person exposure networks [13], documents an efficiency-mixing tension in local consumption [1], and shows that replacing self-reported income with home-neighborhood income can mechanically lower measured mixing [8]. Chetty et al. separate exposure from tie formation conditional on exposure [5]. Chicago's public TNP data contain neither stable persons nor verified co-presence, and ACS income is a neighborhood attribute rather than rider income. We therefore estimate admissible-set opportunity exposure, not realized contact, preference homophily, or an echo chamber.

*Point linkage is not our output.* Sadinle treats record linkage as globally constrained bipartite matching and quantifies uncertainty over admissible links [16]. Our setting has one event file, no observed co-rider edge, and weak labels on nodes rather than labeled record pairs. Physical constraints define the untrimmed feasible set; weak supervision supplies only a declared score restriction; and

**Table 1: Scientific content of the privacy-coarsened release.**

| Released object            | What it identifies                                      | What it does not identify                            |
|----------------------------|---------------------------------------------------------|------------------------------------------------------|
| Match indicator            | Row entered a successful shared service chain           | Partner, group, vehicle, or overlap in occupancy     |
| trips_pooled               | Customer-trip count from empty vehicle to empty vehicle | Common chain key, order, or simultaneous presence    |
| Rounded time and centroids | A declared physical candidate set                       | Realized assignment within that set                  |
| Origin ACS join            | Neighborhood socioeconomic context                      | Rider income, race, or preference                    |
| Missing stable IDs         | Structural absence of repeated persons and groups       | Durable ties, belief reinforcement, or echo chambers |

optimization propagates the surviving ambiguity into the downstream exposure functional. AI is useful only when it tightens that set without defeating known-truth coverage, not because it produces a point-imputed graph.

## 3 DATA AND SCIENTIFIC SETTING

### 3.1 Chicago TNP records

The City of Chicago's 2025- TNP release contains one record per customer trip [6] and includes start and end timestamps rounded to 15 minutes, trip distance and duration, pickup and drop-off census tracts or community areas, centroid coordinates, fares and charges, plus authorization, match, and pooled-chain-size fields. The last field, `trips_pooled`, counts all customer trips from the time the vehicle was empty until it became empty again; two component riders need not have been simultaneously present. Each component has a separate row and the same count, but the common chain ID is suppressed. Accordingly, our pair estimand is a service-chain compatibility opportunity, not physical co-presence.

### 3.2 What the observation operator preserves

The release does not merely add noise to a conventional supervised-learning table. It preserves outcomes on individual rows while deleting the relational key that would join those rows. Table 1 separates what each field licenses from the stronger interpretation it cannot support. This distinction fixes the analysis unit before modeling: trips are observed nodes, two-trip service chains are latent exact-cover pairs, and every reported exposure statistic is conditional on a declared set of feasible joins.

Two sources of selection remain visible even before latent pairing. Sharing authorization determines who can enter the pool, and geographic suppression determines whose neighborhood context can enter  $H$  or  $D$ . We therefore keep authorization/matching outcomes on their full eligible denominators, restrict the primary exposure statistic to pair-eligible rows with joined origin geography, and report the resulting support loss by treatment cell. Neither complete-case analysis nor a learned score repairs non-random suppression; both only define a transparent target population.

A deterministic two-character trip-ID-prefix sample covers December 9, 2025 through February 2, 2026 and contains 53,241 trips,

**Table 2: Completed audit of the Chicago prefix extract.**

| Quantity                                | Measured value  |
|-----------------------------------------|-----------------|
| Trips                                   | 53,241          |
| Shared option authorized                | 3,048 (5.72%)   |
| Matched / all trips                     | 2,146 (4.03%)   |
| Matched / authorized                    | 70.41%          |
| Matched without authorization           | 0               |
| Matched with trips_pooled < 2           | 83              |
| Missing pickup / drop-off tract         | 38.05% / 38.39% |
| Missing pickup / drop-off centroid      | 8.98% / 9.21%   |
| ACS join given released tract (PU / DO) | 95.24% / 94.99% |

*Note.* Measured in the frozen 53,241-row identifier-prefix extract; not population estimates.

including 3,048 authorized and 2,146 matched records. It is suitable for schema tests and aggregate-rate feasibility, but not latent-edge reconstruction: conditional on one component being sampled, its counterpart is omitted with probability approximately 255/256 under the sampling device. Structural estimation therefore requires complete narrow day slices. The completed audit finds 70.41% matched among authorized trips, zero matches without authorization, and 83 matched records with `trips_pooled` < 2. Pickup tracts and centroids are missing for 38.05% and 8.98% of rows; the corresponding drop-off rates are 38.39% and 9.21%. Conditional on release, 95.24% of pickup and 94.99% of drop-off tracts join to the ACS file. These quantities describe this prefix extract, not the Chicago population.

### 3.3 Socioeconomic proxies

We join pickup census tracts to the 2024 five-year American Community Survey Detailed Tables B19013 and B03002 [21]. The prepared extract contains 1,551 Chicago-area tracts and supplies median household income and race/ethnicity composition. Income and composition refer to the trip’s origin neighborhood, not to the rider. The primary pair statistic uses ACS income quintiles; continuous sensitivity analyses use absolute log income differences. Rows without a released and joined pickup tract remain in match-outcome estimation but not in the primary SES bound.

### 3.4 Policy setting and completed feasibility signal

Chicago changed its congestion-zone surcharge structure on January 6, 2026 [7]. Inside the new zone, the solo-shared surcharge gap is \$0.90 during weekday active hours and \$1.50 on weekends. Our planned estimand is the policy intention-to-treat effect on authorization, successful pooling, and the opportunity-exposure interval, using a post-by-zone-by-active-hours design. The policy changes both market thickness and selection into the shared pool, so thickness is a mechanism, not an excluded instrument.

In the prefix feasibility sample, the approximate newly covered zone during weekday active hours shows authorization increasing from 5.95% to 8.08% and matched/all trips from 4.61% to 6.89%. Mean additional charges increase from \$5.69 to \$6.31 in the treated cell. These are unadjusted go/no-go signals, not causal estimates. Complete-day rows, not this prefix, are required for the candidate

graph and policy contrasts. Appendix B records the raw feasibility cells, frozen extraction protocol, data status, and an NYC temporal-coarsening stress test.

## 4 WEAKLY SUPERVISED STRUCTURED INFERENCE

### 4.1 Target: exposure over latent pairings

The target is a statistic of feasible pairings, not a reconstructed social network. Let  $V = \{1, \dots, n\}$  index authorized trips in a complete market-day slice. Trip  $i$  has released features  $x_i$ , node match label  $y_i \in \{0, 1\}$ , released service-chain size  $k_i$ , pickup-neighborhood log income  $Z_i$ , and predeclared income bin  $B_i$ . The suppressed object is a set  $M$  of mutually non-conflicting pairs. For the primary pair analysis, we retain  $V_1 = \{i : y_i = 1, k_i = 2\}$  with valid geography and require every retained node to appear exactly once. If  $m = |V_1|/2$ , the target statistics are

$$\begin{aligned}
 H(M) &= m^{-1} \sum_{\{i,j\} \in M} \mathbb{I}\{B_i = B_j\}, \\
 D(M) &= m^{-1} \sum_{\{i,j\} \in M} |Z_i - Z_j|.
 \end{aligned} \tag{1}$$

$H$  is the share of pairs from the same pickup-neighborhood income bin, not degree-corrected network assortativity;  $D$  is a continuous sensitivity outcome. Both describe neighborhood-level opportunity exposure. Neither the pairing nor verified co-presence is observed.

### 4.2 Feasibility: candidate construction

A candidate edge says only that two records could have belonged to the same two-trip service chain; it is not an inferred co-rider link. We admit an undirected edge  $(i, j)$  only within a calendar day and train/test split when the start-time gap is at most 30 minutes, pickup and drop-off distances are at most 4 and 7 km, and OD-vector cosine is at least  $-0.25$ . A normalized spatiotemporal nearest-neighbor query proposes at most 96 records. Physical cost ranks these proposals, and a deterministic cap limits degree to 16. The surviving edges form  $E_C$ .

The production-primary feature map  $\phi_{ij}$  has 22 columns covering time, OD, duration, mileage, direction, interval overlap, squares, exponentials, and predeclared interactions. A 28-column diagnostic map adds six same-tract/same-area indicators and interactions. Neither map uses endpoint match status, `trips_pooled`, fare, or ACS SES. Search and degree caps can delete the true edge, so every result is conditional on  $E_C$ . We report support, degree, and threshold sensitivity; true-edge recall is observable only in synthetic validation.

### 4.3 Likelihood: node-label weak supervision

The public label says whether a trip matched, not which incident edge was realized. We therefore learn edge compatibility only through a bag-level node likelihood. For candidate edge  $(i, j)$ , let  $a_{ij} = \phi_{ij}^\top \theta$  and  $q_{ij} = \sigma(a_{ij})$ . Incident edges form a bag for node  $i$



and induce the noisy-OR probability

$$\begin{aligned}\hat{p}_i(\theta) &= 1 - \prod_{j:(i,j) \in E_C} (1 - q_{ij}) \\ &= 1 - \exp \left\{ - \sum_{j:(i,j) \in E_C} \text{softplus}(a_{ij}) \right\}.\end{aligned}\quad (2)$$

On candidate-supported training nodes  $\mathcal{T}$ , we estimate

$$\hat{\theta} \in \arg \min_{\theta} \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \{-y_i \log \hat{p}_i - (1 - y_i) \log(1 - \hat{p}_i)\} + \frac{\lambda}{2} \|\theta_{-0}\|_2^2, \quad (3)$$

with  $\lambda = 10^{-3}$ , an unpenalized intercept, stable softplus evaluation, and L-BFGS-B optimization. The fitted  $q_{ij}$  is a compatibility score inside this working likelihood. It is neither an observed edge label nor a calibrated posterior probability that two riders shared a vehicle.

The learned model is linear over the fixed nonlinear basis above. A transparent rule uses the same physical dissimilarities with disclosed fixed weights; only its intercept and nonnegative global scale are fitted by Eq. (3). Whole days form the train/test split. The likelihood therefore supports node prediction only. Public-data pair accuracy is undefined; synthetic truth is used only after fitting to audit candidate recall and within-node edge ranks.

#### 4.4 Optimization: exact-cover exposure bounds

The scientific output is the range of  $H$  or  $D$  over all exact-cover pairings consistent with the declared graph, not the model's highest-scoring pairing. For candidate edge  $e$ , let  $z_e \in \{0, 1\}$  indicate selection, let  $w_e \geq 0$  equal the fitted  $q_e$  (or the comparator's rule score), and let  $g_e$  be an SES statistic such as  $\mathbb{I}\{B_i = B_j\}$  or  $|Z_i - Z_j|$ . The admissible exact covers are

$$\mathcal{M}(E_C) = \{z \in \{0, 1\}^{|E_C|} : \sum_{e \ni i} z_e = 1 \quad \forall i \in V_1, \quad z_e = 0 \quad \text{if } e \notin V_1\}. \quad (4)$$

Every feasible solution selects  $|V_1|/2$  edges, so the untrimmed admissible interval is

$$\left[ \frac{2}{|V_1|} \min_{z \in \mathcal{M}(E_C)} \sum_e g_e z_e, \frac{2}{|V_1|} \max_{z \in \mathcal{M}(E_C)} \sum_e g_e z_e \right]. \quad (5)$$

AI enters only as a declared sensitivity restriction. Let  $W^* = \max_{z \in \mathcal{M}(E_C)} \sum_e w_e z_e$  and define

$$\mathcal{M}_\rho(E_C) = \left\{ z \in \mathcal{M}(E_C) : \sum_e w_e z_e \geq \rho W^* \right\}. \quad (6)$$

Replacing  $\mathcal{M}(E_C)$  by  $\mathcal{M}_\rho(E_C)$  in Eq. (5) yields a score-restricted interval. Here  $\rho$  is an analyst-chosen sensitivity parameter, not a confidence level, and the resulting interval is not a confidence interval. All programs are binary linear programs solved with HiGHS and reported only with a global optimality certificate.

**PROPOSITION 1 (CONDITIONAL SHARPNESS AND COVERAGE).** *If  $\mathcal{M}(E_C)$  is nonempty and both extrema are solved globally, Eq. (5) is sharp over the declared candidate graph. If the suppressed true pairing  $z^\dagger$  belongs to  $\mathcal{M}(E_C)$ , the untrimmed interval contains its target value. The same membership-based guarantee holds for a score-restricted interval exactly when  $z^\dagger \in \mathcal{M}_\rho(E_C)$ , equivalently  $\sum_e w_e z_e^\dagger \geq \rho W^*$ .*

Both feasible sets are finite, so certified extrema are attained and bracket every member. Moreover,  $\mathcal{M}_{\rho_2}(E_C) \subseteq \mathcal{M}_{\rho_1}(E_C)$  for  $\rho_2 \geq \rho_1$ : increasing  $\rho$  can narrow the interval, but it can also discard the true pairing. Failure of the score floor removes the structural coverage guarantee; it need not numerically exclude the same scalar target. The appendix gives the formal proof.

#### 4.5 Interpretation and audit logic

The construction compares two nested admissible sets. Conditional on the exact-cover cohort and field consistency,  $\mathcal{M}(E_C)$  additionally requires the physical screen to contain every retained true pair. It does not assume that a learned score ranks that pair highly. The restricted set  $\mathcal{M}_\rho(E_C)$  adds one claim: an admissible pairing must retain at least a declared fraction of the best total compatibility score. Reporting both sets makes the source of sharpening visible. If the raw interval is wide, the public observation operator leaves substantial structural ambiguity. If only the score-restricted interval is narrow, the apparent information comes from the model and must survive a known-truth coverage audit.

Exact-cover optimization is essential because edge decisions are coupled. A high-scoring edge consumes both endpoints, changes which other edges remain available, and changes the composition of a complete pairing. Thresholding edges independently, clustering by proximity, or taking the maximum-score pairing would answer a different question: which single graph an algorithm prefers. Our min/max programs instead ask which exposure values remain possible after every retained matched node is assigned exactly once. They can return a wide range or infeasibility, and either outcome is part of the scientific result.

The validation targets the sequence of claims created by this construction. Candidate recall tests whether truth enters  $E_C$ . Held-out Brier loss tests whether edge scores aggregate to calibrated node matchability. Hidden-edge ranking tests whether good node prediction orders the unobserved relation in a known-truth market. Finally, true-pairing eligibility tests whether the score floor preserves the member whose target generated the synthetic data. Only the first and last checks support structural coverage; node prediction alone does not. In Chicago, where pair truth is unavailable, we can report node calibration, support, feasibility, and sensitivity across declared graphs, but we cannot turn those diagnostics into empirical pair accuracy.

These intervals also differ from sampling confidence intervals. The admissible range conditions on one released market slice and varies the suppressed relation. Sampling uncertainty instead varies market slices or dates. The complete analysis will recompute structural endpoints inside date-block resamples and will display the untrimmed range, the family indexed by  $\rho$ , and sampling uncertainty separately. No choice of  $\rho$  is calibrated to have 90% or 95% frequentist coverage.

### 5 IDENTIFICATION CONDITIONS AND FALSIFICATION

Proposition 1 is deliberately conditional. It does not turn privacy-coarsened records into observed relationships; it states what follows after the candidate graph, exact-cover cohort, and optional score floor are declared. Table 3 makes those conditions operational and

preassigns the interpretation of their failure. This prevents a favorable point estimate or narrow interval from overruling a failed measurement check.

Candidate completeness is the strongest structural requirement and the least directly testable in Chicago. Search windows and degree caps make computation possible, but either can delete a true pair. High node support is therefore not proof of high true-edge recall: a node can have many physically plausible neighbors while its actual service-chain partner is absent. We use known-truth markets to measure recall and use Chicago support, component structure, and threshold sensitivity to reveal fragile regions. The resulting raw range is sharp over  $E_C$ , not over every relationship that could exist under a looser physical model.

Exact-cover feasibility provides a different diagnostic. An infeasible component can reflect an overly narrow candidate rule, inconsistent public chain-size fields, or eligibility decisions that remove a required counterpart. It is not repaired by matching the largest feasible subset, because doing so changes both the population and the exposure denominator. Production results therefore report infeasibility by date and socioeconomic support stratum. Any revised candidate graph is run as a separately declared sensitivity analysis, not chosen because its exposure estimate is more precise.

Score alignment is an additional assumption, not a consequence of feasibility. The noisy-OR likelihood rewards node-level prediction after aggregating all incident edges. It can succeed by learning candidate degree, market density, or a geographic shortcut while ordering the true partner poorly. This is why the validation reports Brier loss, hidden-edge rank, and true-pairing eligibility as distinct outcomes. If a score restriction fails the known-truth audit, a narrow range is evidence about the model's preference, not stronger identification.

Finally,  $H$  and  $D$  inherit the meaning of their contextual labels. They compare released pickup-neighborhood income, not rider resources or preferences. A high same-bin share may reflect residential sorting, trip geography, platform assignment, or chance within a constrained feasible set. Without persons, repeated ties, or beliefs, it cannot establish preference homophily or an echo chamber. The urban contribution is narrower: it asks whether the algorithmically mediated opportunity set is compatible with more or less cross-neighborhood exposure under explicit observation and model restrictions.

## 6 VALIDATION: SHARPENING AND FALSE PRECISION

Before the first held-out run, we fixed a two-day Chicago-coordinate generator with 80 hidden pairs and 80 unmatched authorized trips per day, 15-minute time rounding, model defaults, and score retentions  $\rho \in \{0.90, 0.95\}$ . Pair IDs and synthetic income bins were stored in separate truth files and never read by candidate construction, model fitting, or score-restricted optimization. Day one is used for fitting; day two is evaluated once.

*Candidate support comes first.* The holdout graph retains all 80 hidden pairs. This check is logically prior to prediction: if a true pair were absent, even the untrimmed interval would lose the coverage

guarantee in Proposition 1. Across a separate 20-replicate implementation check, candidate recall and untrimmed scalar coverage are 100% at 1-, 5-, 15-, and 30-minute rounding; Appendix B reports the full coarsening table.

*Node calibration does not imply edge recovery.* On 240 candidate-supported holdout nodes, the rule has Brier loss 0.06048, compared with 0.00595 for the 22-feature primary score and 0.00513 for the 28-feature diagnostic score. Both learned scores therefore look excellent on the objective they were trained to predict. Hidden-edge ranks separate them. The primary score raises mean reciprocal rank from 0.781 to 0.941 and top-1 accuracy from 67.5% to 90.0%. The diagnostic score instead falls to 0.775 and 64.4%. Better node prediction is not evidence of better pair recovery; no edge label enters training.

### Eligibility and containment, not width alone, audit sharpening.

The hidden same-income-bin share is 0.5625. Physical admissibility alone gives  $[0.050, 0.775]$ . Score retention narrows this set, but Table 4 shows that width cannot be read without coverage. The primary score covers the hidden value and retains the complete hidden pairing at both score floors. The diagnostic  $\rho = 0.95$  region is only 0.162 wide but lies above the hidden value and excludes the hidden pairing. Even scalar containment is a weak check: the rule's  $\rho = 0.95$  interval happens to contain 0.5625 although the complete hidden pairing fails its score floor.

*Geography equality manufactures precision.* The diagnostic map adds six same-tract and same-area columns without changing a single candidate edge: both models optimize over the same 2,640-edge graph. Among the 560 candidates incident to matched holdout nodes, pickup- and drop-off-tract equality coincide exactly with the synthetic same-income-bin target, while area equality is constant. The diagnostic model can therefore improve its node objective while sorting pairings by a proxy for the evaluation outcome. Removing those columns worsens node Brier only from 0.00513 to 0.00595, improves edge ranking, and restores coverage. This is feature-induced false precision, not proof that the primary map is proxy-free: continuous coordinates may still encode socioeconomic geography.

At  $\rho = 0.90$ , the primary interval is slightly wider than the rule interval (0.475 versus 0.438); at  $\rho = 0.95$  it is slightly narrower (0.300 versus 0.312) and uniquely retains the hidden pairing. The pilot validates the coverage audit and the possibility of safe sharpening. It does not establish that AI uniformly tightens a transparent rule.

*The prefix extract is a negative software check.* The incomplete Chicago prefix graph contains 151 edges and supports only 246 of 3,048 authorized nodes (8.1%). On 93 supported holdout nodes, weak MIL has Brier loss 0.15777 versus 0.15525 for the rule. Under approximately independent prefix retention, a two-trip chain's other component is absent with probability 255/256. We therefore neither tune the graph nor run exact-cover bounds on this extract. Complete-day Chicago support, calibration, exposure ranges, and policy contrasts remain reserved for the predeclared extraction.

**Table 3: Identification conditions, diagnostics, and retained claims.**

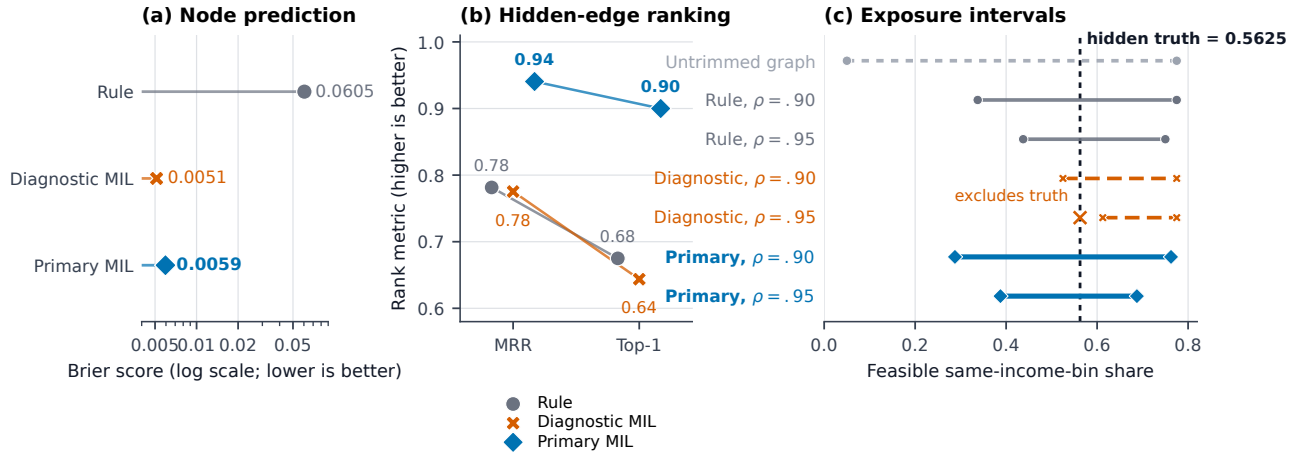
| Condition                    | Evidence reported                                                                             | Interpretation if the condition fails                                        |
|------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Candidate completeness       | Known-truth recall; node support, degree, component size, and threshold grid in Chicago       | Untrimmed range is conditional only; no structural coverage claim            |
| Exact-cover consistency      | Parity, feasibility, solver certificate, and enumeration agreement on small graphs            | Cell is infeasible; do not drop nodes or substitute a best-score graph       |
| Score alignment              | Out-of-day node calibration plus hidden-edge rank and true-pairing eligibility in simulation  | Learned restriction is diagnostic; rule and untrimmed ranges remain primary  |
| Contextual target validity   | ACS release/join rates, suppression strata, continuous-gap and area-level alternatives        | Claim is limited to the observed neighborhood-context sub-population         |
| Policy measurement stability | Charge first stage, pre-trends, support continuity, and sign stability across declared graphs | Report measurement shift, descriptive association, or unidentified direction |

**Table 4: Locked holdout: narrowing, scalar containment, and structural eligibility.**

| Admissible set               | Interval       | Width | Reduction | Contains value | True pairing eligible |
|------------------------------|----------------|-------|-----------|----------------|-----------------------|
| Untrimmed candidate graph    | [0.050, 0.775] | 0.725 | —         | Yes            | Yes                   |
| Rule, $\rho = 0.90$          | [0.338, 0.775] | 0.438 | 39.7%     | Yes            | Yes                   |
| Primary AI, $\rho = 0.90$    | [0.288, 0.763] | 0.475 | 34.5%     | Yes            | Yes                   |
| Rule, $\rho = 0.95$          | [0.438, 0.750] | 0.312 | 56.9%     | Yes            | No                    |
| Primary AI, $\rho = 0.95$    | [0.388, 0.688] | 0.300 | 58.6%     | Yes            | Yes                   |
| Diagnostic AI, $\rho = 0.90$ | [0.525, 0.775] | 0.250 | 65.5%     | Yes            | Yes                   |
| Diagnostic AI, $\rho = 0.95$ | [0.613, 0.775] | 0.162 | 77.6%     | No             | No                    |

Note. Hidden target = 0.5625. Reduction is relative to the untrimmed range. “Eligible” means that the complete hidden pairing satisfies the score floor in Proposition 1; numerical containment of its scalar target is weaker. These are sensitivity regions, not confidence intervals.

### Known-truth synthetic benchmark (not an empirical Chicago estimate)



**Figure 1: The locked benchmark separates three claims. Candidate recall is 80/80 before scoring. Both learned models improve held-out node Brier loss, but only the geography-equality-free primary model improves hidden-edge ranking. The shortcut-rich diagnostic model produces the narrowest  $\rho = 0.95$  region and excludes hidden truth. The dashed line and hidden pairs are never supplied to fitting.**

## 7 COMPLETE-DAY POLICY DESIGN

Chicago changed its congestion-zone surcharge on January 6, 2026. We estimate an intention-to-treat contrast comparing post-policy trips in the covered zone during active hours with the corresponding spatial and temporal controls. Outcomes are authorization among

eligible trips and successful pooling among all eligible trips. The surcharge can change both the size and composition of the shared pool; market thickness is therefore a mechanism, not an excluded instrument. Event-time leads and the charge discontinuity must validate timing and pre-trends before causal language is used.

The complete-day analysis distinguishes a volume response from a composition response. Authorization/all trips measures entry into the eligible shared pool; matched/all trips measures successful pooling without conditioning on that post-policy choice. Candidate support and exact-cover feasibility are then audited on the same spatial and temporal cells used for exposure. A discontinuity in supported composition at the policy boundary would be a measurement change, not evidence of mixing. The design therefore requires stable support and parallel pre-trends in addition to a visible surcharge first stage.

For opportunity exposure, regressing estimated daily lower endpoints does not generally produce a lower bound on a policy coefficient. Let  $\mathcal{F}_c$  denote the admissible pairings in market-time block  $c$  and let  $a_c$  be the fixed weights of the policy contrast. We propagate latent-edge ambiguity through the contrast itself:

$$\begin{aligned}\underline{\theta} &= \min_{M_c \in \mathcal{F}_c} \sum_c a_c H(M_c), \\ \bar{\theta} &= \max_{M_c \in \mathcal{F}_c} \sum_c a_c H(M_c).\end{aligned}\tag{7}$$

Date-block resampling layers sampling uncertainty on this structural range while holding the candidate rule fixed.

Equation (7) prevents an invalid shortcut. The lower endpoint on each date can be produced by a different latent pairing, and the weights  $a_c$  can be positive or negative. Regressing a sequence of daily lower endpoints therefore need not minimize the policy contrast. Optimizing the contrast jointly over all block-specific pairings preserves the declared admissible set. If this joint range crosses zero, the direction of the exposure effect is not identified even when one point reconstruction has a stable sign.

The analysis has three falsifiable hypotheses. **H1 (thickness)** predicts a positive ITT on authorization/all and matched/all. **H2 (opportunity exposure)** is two-sided: the policy range in Eq. (7) may move toward cross-neighborhood exposure or toward geographically segmented pooling. Its result is the sign of the full range, not the sign of one imputed network. **H3 (AI value)** is a complete-day gate: relative to the rule at the same score floor, the learned score must improve held-out node Brier loss by at least 10% and reduce interval width by at least 25% without changing the conclusion across the declared candidate grid. The synthetic benchmark audits this gate and its failure mode; it does not claim that H3 has been met. Failure of H3 leaves the rule and untrimmed ranges primary and does not invalidate H1.

## 8 DISCUSSION

The urban object here is neither residential segregation nor a realized social network. It is the set of cross-neighborhood service-chain opportunities that could have generated the public records. This object matters because a pooling algorithm can alter who is combinatorially available to whom before any durable tie or attitudinal change occurs. A thicker market may widen that set across neighborhoods, or it may add feasible pairs inside geographically segmented submarkets. The direction is an empirical property of the admissible exposure range, not a consequence of higher match volume.

The methodological distinction is the treatment of privacy suppression as a known observation operator on relations. Conventional node prediction leaves the scientific outcome undefined; point linkage replaces missingness with one untestable graph. Here physical restrictions define a score-free set, and weak supervision can only remove members through an explicit sensitivity rule. This turns an otherwise hidden modeling choice into an auditable comparison between the untrimmed and restricted ranges. The same pattern applies whenever public data preserve object-level outcomes but suppress groups, assignments, or encounters.

The locked failure is as important as the locked gain. Geography equality improves the trained node objective and produces a much narrower interval, yet at  $\rho = 0.95$  it removes the true pairing and the true target. Thus predictive fit cannot certify relational identification. The useful AI contribution is conditional sharpening with visible coverage cost: candidate support, node calibration, hidden-edge ranking where truth exists, and structural eligibility must be read together. Without complete-day Chicago data, this remains a method and audit result rather than a finding about homophily or urban mixing.

### 8.1 Implications for urban AI and market design

The AI4Sciences role is not to replace a hidden graph with a more elaborate prediction. It is to state how a learned restriction changes a scientific identified set. The observation operator, physical candidate rule, node likelihood, score floor, and exposure objective are separate objects; each can be inspected or varied without silently redefining the others. This modularity creates a falsifiable AI contribution. A model adds information only when its restricted range is narrower, its predictive diagnostics improve out of day, and known-truth audits do not show that the narrowing comes from excluding the data-generating relation. Otherwise the scientifically relevant output is the score-free range or a declaration of non-identification.

This framing also separates two operations questions that are often collapsed. Market thickness concerns how many requests are available and whether more of them match. Opportunity composition concerns which feasible assignments account for those outcomes. A policy can raise authorization and successful pooling while leaving a wide exposure range, or while shifting every admissible pairing toward more same-neighborhood opportunities. Conversely, match volume may change little while the composition of feasible pairs changes. The policy ITT therefore applies to both entry and assignment opportunity; thickness is a measured mechanism, not an exclusion restriction that isolates dispatch.

The method produces an evidence hierarchy for the eventual Chicago analysis. A one-signed untrimmed policy range would be the strongest result available from the declared candidate graph because it does not depend on learned edge weights. A one-signed restricted range paired with an untrimmed range that crosses zero would be conditional evidence: the conclusion follows only for pairings sufficiently compatible with the stated score and  $\rho$ . Agreement between the rule and primary model would show robustness to two score maps, not truth of either map. Disagreement, loss of



true-pairing eligibility in simulation, or sign changes across candidate graphs would downgrade the claim to model sensitivity. These categories are fixed before the policy outcome is observed.

There is also a practical distinction between auditing and optimization. The reported interval can reveal that a platform’s public outcomes are compatible with sharply segmented or broadly mixed opportunities. It does not identify the proprietary objective, and it does not recommend forcing cross-neighborhood pairs. An operational intervention would also have to account for detour, waiting, reliability, price, driver burden, and rider choice—quantities absent from the released table. The admissible-set analysis is therefore an input to oversight, not a standalone welfare objective.

## 8.2 Transfer and scope

The same structure appears when administrative data preserve item-level outcomes but suppress teams, referral chains, shipments, or shared resources. Transfer requires a domain-specific feasibility screen and group constraint; the ride-pooling thresholds are not portable. The general lesson is narrower: when the missing scientific object is relational, validation should test candidate support and downstream set coverage, not only node prediction. A learned score may sharpen the answer, but the score-free feasible set must remain visible as the reference against which that sharpening is judged.

## 9 LIMITATIONS AND ETHICAL CONSIDERATIONS

This study uses public, privacy-coarsened administrative records and aggregate census statistics; it recruits no participants and performs no intervention. The applicable institutional determination will be obtained and disclosed before submission. Neighborhood income is not rider income, a candidate edge is not verified co-presence, and trips\_pooled can join components of one service chain that were never simultaneously occupied. Without a stable rider identifier, the data cannot measure durable homophily, repeated exposure, belief reinforcement, or an echo chamber. Non-random geographic suppression, timestamp rounding, and ACS join failure can also change the analyzed subpopulation and are reported by treatment cell.

Publishing one reconstructed graph could expose identities or stigmatize neighborhoods. Released artifacts therefore contain aggregate intervals, synthetic examples, and code, but no row-level inferred social graph. Score restrictions can under-cover, so the untrimmed range and sensitivity grid remain visible. The bounds are an auditing device, not a license to infer sensitive attributes or optimize mixing at the expense of wait, detour, surveillance, or disparate service burdens. If the effect range overlaps zero or changes sign across declared graphs, the conclusion is non-identification.

## 10 GENERATIVE AI USAGE

Generative AI tools (OpenAI ChatGPT and Codex; exact model identifiers and access dates to be completed at submission) were used to assist code scaffolding and debugging, literature-search organization, prose drafting, and formatting. All research questions, estimands, model restrictions, analyses, source verification, and claims remain the responsibility of the authors. AI-generated code was executed against deterministic tests and known-truth

simulations; numerical results in the paper are produced by saved scripts rather than generated text. No confidential participant data or proprietary platform records were supplied to a generative AI system for this draft. Generative AI did not recruit or interact with participants, label empirical observations, or appear as an author. The final disclosure will enumerate every material use across the full research process.

## 11 CONCLUSION

The missing object in privacy-coarsened pooling records is the relation, not another trip feature. BOUNDPOOL therefore reports an admissible exposure interval rather than one reconstructed graph: physical constraints define the exact-cover set, weak supervision supplies an optional score restriction, and optimization carries the surviving ambiguity into the estimand. The locked benchmark shows why AI is a tightening device rather than a discovery device. A geography-equality-free score can narrow the range while retaining the hidden pairing, whereas a shortcut-rich score looks sharper and excludes truth. The complete-day Chicago intervals and policy sign remain reserved. This pilot is evidence for the audit, not evidence that ride pooling changes urban social mixing.



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## A REPRODUCIBILITY APPENDIX

The artifact contains the complete candidate builder, weak noisy-OR model, MILP bound solver, locked synthetic generator, design lock, unit tests, and machine-readable outputs. The current implementation uses NumPy, pandas, SciPy/HiGHS, scikit-learn, and Matplotlib. Seven exact-cover tests cover exact matching, infeasibility, score restriction, SES gaps, unmatched-node exclusion, synthetic candidate recall, and agreement between the MILP and an exhaustive small-graph fallback.

The complete-day extraction requests every authorized row for two comparable weekdays, checks server-side counts, writes atomically, hashes files, and audits duplicate IDs, window boundaries, match/pooling consistency, and spatial missingness. The current execution environment blocked both supported City API routes; no row count or partial complete-day file is reported in this draft.

## B AUXILIARY DIAGNOSTICS AND FROZEN EMPIRICAL PROTOCOL

## B.1 Known-truth diagnostics

Table 5: Locked synthetic holdout: node and edge diagnostics.

| Metric           | Rule    | Primary AI     | Diagnostic AI |
|------------------|---------|----------------|---------------|
| Node Brier loss  | 0.06048 | <b>0.00595</b> | 0.00513       |
| Node log loss    | 0.25443 | <b>0.04732</b> | 0.04206       |
| Node ROC–AUC     | 0.9603  | <b>1.0000</b>  | 1.0000        |
| True-edge MRR    | 0.781   | <b>0.941</b>   | 0.775         |
| True edge top-1  | 67.5%   | <b>90.0%</b>   | 64.4%         |
| True edge top-3  | 85.6%   | <b>98.1%</b>   | 90.0%         |
| Candidate recall |         | 80/80 (100%)   |               |

Note. Ranking conditions on the hidden true edge being in the candidate graph; no edge labels enter training. Primary AI removes six geography-equality columns; diagnostic AI retains all 28 columns.

Across 20 independent small-market replications, candidate recall and untrimmed coverage equal 100% for 1-, 5-, 15-, and 30-minute rounding. The edge/true-edge ratio rises from 1.04 to 2.11 and mean raw width from 0.003 to 0.150 as time is coarsened. Under the deliberately well-specified  $\rho = 0.95$  rule restriction, mean width at 15 minutes falls to 0.020 and at 30 minutes to 0.027 with 100% scalar coverage. This is an implementation check, not evidence that the Chicago candidate model is correctly specified.

## B.2 Data status and raw feasibility cells

The production downloader requests a server-side count for each complete day, retrieves every authorized row in deterministic identifier order, writes atomically, and records the query, retrieval time,

**Table 6: Candidate/solver check over 20 replicates.**

| Bin<br>(min) | Edge /<br>truth | Raw<br>width | $\rho = .95$<br>width | Width<br>reduction |
|--------------|-----------------|--------------|-----------------------|--------------------|
| 1            | 1.038           | 0.003        | 0.003                 | 0.0%               |
| 5            | 1.197           | 0.020        | 0.007                 | 66.7%              |
| 15           | 1.563           | 0.073        | 0.020                 | 72.7%              |
| 30           | 2.107           | 0.150        | 0.027                 | 82.2%              |

Note. Candidate recall, untrimmed scalar coverage, and score-restricted scalar coverage equal 100% in every row.

**Table 7: Data layers and current scientific use.**

| Layer                   | Status        | Scientific use                                  |
|-------------------------|---------------|-------------------------------------------------|
| Chicago prefix sample   | Complete      | Schema, missingness, aggregate feasibility only |
| Chicago complete days   | Reserved      | Candidate graph, node calibration, SES bounds   |
| ACS 2024 tracts         | Complete      | Pickup-neighborhood SES proxy                   |
| NYC HVFHV               | Pilot extract | Coarsening stress test; no pair truth           |
| Locked synthetic market | Complete      | Recall, ranking, coverage audit                 |

**Table 8: Raw policy-window feasibility signals (not causal).**

| Cell       | Period | $N$    | Authorized | Matched/all | Add'l charges |
|------------|--------|--------|------------|-------------|---------------|
| Target     | Pre    | 3,797  | 5.95%      | 4.61%       | \$5.69        |
| Target     | Post   | 4,047  | 8.08%      | 6.89%       | \$6.31        |
| Comparison | Pre    | 10,975 | 5.53%      | 3.54%       | \$6.86        |
| Comparison | Post   | 11,369 | 5.82%      | 3.97%       | \$6.85        |

Note. Four-week windows around January 6, 2026; approximate feasibility-stage zone; no uncertainty or causal adjustment.

**Table 9: Predeclared complete-day empirical gates.**

| Gate                                       | Threshold            | Result   |
|--------------------------------------------|----------------------|----------|
| Held-out Brier improvement vs. rule        | $\geq 10\%$          | Reserved |
| Candidate-supported matched nodes          | report, no target    | Reserved |
| AI width reduction vs. rule at same $\rho$ | $\geq 25\%$          | Reserved |
| Exact-cover feasibility                    | $\geq 95\%$ of cells | Reserved |
| Sign stability across candidate grid       | same conclusion      | Reserved |
| Post-policy observation window             | $\geq 8$ months      | Reserved |

row count, and SHA-256 digest. Audits cover identifiers, date boundaries, boolean consistency, service-chain codes, coordinates, and tract suppression. Partial downloads are never combined with complete days. Days, not edges, form train/test units; ACS joins occur after candidate support is frozen, and SES never enters candidate construction or score fitting. Every result carries its eligible and supported denominators, degree distribution, infeasibility rate, and configuration.

The NYC pilot contains 2,000 outcome-stratified HVFHV records with precise times and taxi-zone geography. It supports schema and future 5-, 15-, and 30-minute coarsening checks but cannot estimate match rates or validate pairs: NYC also suppresses the rider, vehicle, and shared-group identifiers.

### B.3 Policy and failure protocol

Let  $Post_t$  indicate trips after January 6,  $Zone_i$  the released spatial proxy for the covered zone, and  $Active_t$  the policy's active hours. The frozen trip-level specification is

$$Y_{ict} = \alpha_c + \delta_t + \beta(Post_t \times Zone_i \times Active_t) + \Gamma' X_{ict} + \varepsilon_{ict}. \quad (8)$$

Event-time coefficients, zone-specific trends, and weekday/weekend dose contrasts accompany Eq. (8). Zone polygons and a boundary ambiguity band are fixed before outcomes are estimated. Diagnostics include lead equivalence tests, placebo dates, alternative spatial cells, holiday and venue exclusions, and the surcharge discontinuity in `additional_charges`. Date-block resampling or a predeclared small-cluster correction is chosen after auditing independent cells.

The candidate grid varies the 30-minute start window to 15 and 45 minutes, pickup radii 4 km to 2 and 6, drop-off radii 7 km to 4 and 10, degree cap 16 to 8 and 32, and  $\rho \in \{0, 0.80, 0.90, 0.95\}$ . No graph is selected after observing exposure. Candidate support, component sizes, degree, exact-cover feasibility, calibration by degree, and geographic suppression are reported by date and treatment cell. Continuous-geometry and ACS-permutation ablations test remaining proxies. Known-truth simulations cross coarsening, route overlap, assortativity, and candidate misspecification; both scalar containment and true pairing eligibility are recorded. A time-limit incumbent is never an endpoint.

Predeclared failures remain results. A partial day stops structural estimation; an infeasible cell stays infeasible; a missed Brier/width gate leaves the rule and untrimmed ranges primary; unstable candidate support is a measurement shift; failed pre-trends imply descriptive association; and a policy-effect range that crosses zero implies an unidentified direction. None is replaced by a maximum-score pairing.

**Table 10: Evidence-to-claim ladder for the completed paper.**

| Claim                    | Required evidence and boundary                                                                        |
|--------------------------|-------------------------------------------------------------------------------------------------------|
| Policy thickened pooling | ITT on authorization/all and matched/all with acceptable pre-trends; not a thickness-only instrument. |
| AI predicts matchability | Out-of-day node calibration beats the rule; does not recover co-rider edges.                          |
| Candidate exposure range | Certified extrema with support and infeasibility audits; conditional on the graph.                    |
| AI-sharpened range       | Score-indexed range, known-truth audit, and explicit $\rho$ ; not a confidence interval.              |
| Policy changed exposure  | One-signed structural range across declared candidates; not dispatch causality or realized contact.   |
| Echo chamber             | Not testable: no repeated ties, beliefs, or reinforcement.                                            |

Mechanism reporting separates authorization by origin-income bin, matched/all by the same bins, the untrimmed physical range, and score-restricted ranges. Composition reweighting is descriptive. Without rejected requests, vehicles, provider identity, or dispatch scores, the public data cannot separate rider selection, supply response, and proprietary assignment.

## C PROOF OF PROPOSITION 1

The feasible binary vectors in  $\mathcal{M}(E_C)$  are in one-to-one correspondence with exact-cover pairings on the declared graph. This set is finite, so certified minimization and maximization attain sharp endpoints and bracket every feasible  $z^\dagger$ . The same argument applies to  $\mathcal{M}_\rho(E_C)$ ; the true pairing remains a member exactly when  $\sum_e w_e z_e^\dagger \geq \rho W^*$ . Finally,  $\rho_2 \geq \rho_1$  implies  $\mathcal{M}_{\rho_2}(E_C) \subseteq \mathcal{M}_{\rho_1}(E_C)$ , which proves the nested-endpoint claim without implying that truth remains feasible or that scalar containment fails whenever membership does.

## D IMPLEMENTATION PSEUDOCODE

### D.1 Degree-capped candidate graph

- (1) Split by date/train-test assignment, project centroids, and build the normalized start-time/pickup-coordinate  $k$ -d tree.

- (2) Query 96 neighbors, apply hard filters, deduplicate by physical cost, and retain at most 16 proposals per node.
- (3) Forbid cross-date/split edges and emit support and degree diagnostics.

### D.2 Out-of-day weak edge scoring

- (1) Construct compatibility features without labels, fares, or SES, then fit the rule intercept/scale on supported training nodes.
- (2) Fit  $\hat{\theta}$  with the node likelihood and no edge label.
- (3) Report held-out calibration; rank edges only in known-truth simulation.

### D.3 Certified exposure range

- (1) Verify graph integrity, finite scores, parity, and feasibility.
- (2) Solve  $W^* = \max_{z \in \mathcal{M}(E_C)} \sum_e w_e z_e$  to global optimality; add the score floor only for a declared  $\rho$  analysis.
- (3) Solve both SES extrema, reject uncertified incumbents, normalize by pair count, and retain certificates/configuration.